

SUBHRANEEL GOSWAMI

☎ +91 6295590363 ✉ subhraneeljobs@gmail.com 🔗 [linkedin.com/in/subhraneel](https://www.linkedin.com/in/subhraneel) 🐙 github.com/subhraneel2005

Education

Institute of Engineering and Management

Bachelor of Computer Applications

Jun. 2023 – May 2026

Kolkata, IN

Experience

Zenera Labs

Aug 2026 – Present

AI Engineer

Bengaluru, India, (Remote)

- Developed and evaluated a **Breast Cancer Classification** model using **Python** and **scikit-learn**, analyzing classification performance across multiple evaluation metrics including **precision, recall, F1-score, and confusion matrix**.
- Taking **end-to-end ownership** of a fully offline, real-time **AI weapon detection inference system** for CCTV feeds, covering model inference, video ingestion, backend services, local deployment, and mobile integration; reviewed asynchronously by the **manager and founder**.
- Engineering an **edge-first detection pipeline** using **YOLOv8, ONNX/NCNN, OpenCV, FastAPI, WebSockets, SQLite, Docker, and Expo/React Native**, enabling local RTSP inference, real-time alerts, detection logs, and mobile monitoring without cloud dependency.

Kasukabe Labs

Oct 2025 – Jan 2026

Fullstack/Applied AI Intern

Kolkata, India, (Remote)

- Contributed to a production **AI video generation pipeline (Python, FastAPI)** architected as **async microservices** for dialogue generation, character design, audio synthesis, and video assembly.
- Built dialogue audio (**LLMs, ElevenLabs**) and character generation (**Gemini, NanoBanana API**) services using **asyncio.gather** for concurrent, rate-limit aware processing.
- Participated in upfront **system design, database schema design**, architecture discussions, **API documentation**, and built **batch-processing workflows** with **rate-limit aware concurrent execution**.

Jobsforce.ai

Mar 2025 – Sep 2025

Full Stack Developer Intern

SF, (Remote)

- Built and shipped a production **Chrome extension** using **Plasmo** under a **one-week deadline**, collaborating closely with a senior engineer after rapidly learning the extension architecture.
- Developed an end-to-end **AI Mock Interviewer** with **React, Node.js, TypeScript, PostgreSQL**, LLM integrations, backend APIs, and real-time interview workflows with **per candidate Report generation**.
- Designed secure website extension **authentication sync**, implemented **WebSocket** based real-time LLM communication, and built scalable frontend data handling for **5,000+ job listings** using pagination and infinite scrolling.

Projects

AI Coding Agent | *TypeScript, Vercel AI SDK, Agentic-Memory, Commander, Ink, Bun*

- Built an **open-source CLI coding agent** with **tool-calling architecture** including **26+ tools** for autonomous filesystem, git, bash, memory tool usage to navigate and modify codebases.
- Designed a **parallel multi-agent orchestrator** where a planner decomposes tasks into file-scoped todos and dispatches isolated executor sub-agents via **Promise.allSettled**, with conflict detection to prevent overlapping writes.
- Designed an **agentic memory architecture** inspired by GitHub Copilot, combining project memory, session context, and persistent knowledge to improve long-running coding workflows.
- Added **human-in-the-loop diff approval** and **path traversal protection** against destructive edits.
- Github: github.com/subhraneel2005/nini

Multi-Agent Customer Support System | *Python, LangChain, LangGraph, FastAPI, Qdrant, Pydantic, Redis, Docker*

- Built a **production-grade multi-agent system** using **LangGraph** with **triage-based routing** across billing, technical support, and human escalation agents.
- Implemented a triage agent for **intent classification** and **entity extraction**, and a **RAG-powered** technical support agent using **Qdrant vector search** with citation-backed, confidence-scored responses.
- Developed a billing agent with **invoice verification**, KB-grounded policy lookup, and structured escalation via **Pydantic** output models.
- Built a **FastAPI REST** surface with **YAML-driven guardrails** and structured **JSON logging** with trace IDs.
- Github: github.com/subhraneel2005/multi-agent-cs-system

Technical Skills

Languages: Python, TypeScript, JavaScript, Go (learning), Java, SQL

Frontend: React, Next.js (App Router, RSC), Expo, React Native, HTML5, CSS3, Tailwind CSS, shadcn/ui

Backend: Node.js, FastAPI, Bun, REST APIs, WebSockets, AsyncIO, Microservices, Server Actions, Authentication, Authorization, JWT, Redis

Databases & ORM: PostgreSQL, MongoDB, SQLite, Prisma

AI/ML: scikit-learn, YOLOv8, ONNX Runtime, NCNN, OpenCV, NumPy, Pandas, Matplotlib, PyTorch

AI & LLM Systems: OpenAI API, Gemini API, Vercel AI SDK, LangGraph, Tool Calling, Streaming, AI Agents, Agentic Memory, RAG, Vector Databases, Qdrant, Sentence Transformers

Infrastructure: Linux, Docker, Docker Compose, AWS, GCP, Vercel, CI/CD, RTSP

Developer Tools: Git, GitHub, Cursor, Claude Code, Opencode